

Feet in the Sand: Unlock the Restorative Power of the Sea through Virtual Activities

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Figure 1: Taputeranga Marine Reserve, 2026.

ABSTRACT

This research explores Mixed Reality (MR) ocean blue spaces to support mental wellbeing among adolescents and young adults, a group vulnerable to mental health disorders [1], when physical ocean access is limited. While natural blue space activities boost health outcomes [2], art-based virtual water experiences remain underexplored for those restricted by location, illness, or mobility. We propose a non-gamified seaside intervention set at Taputeranga Marine Reserve, featuring multi-sensory coastal soundscapes, immersive hand-triggered exploration, guided dance with breathing exercises, and ocean reconnection. Unlike passive VR relaxation, this approach emphasizes active movement, cultural resonance, and place-memory evocation. Cross-cultural workshops planned for Wellington, Aotearoa New Zealand and Busan, South Korea (2026) will evaluate emotional responses, stress relief, and therapeutic potential through multimodal participant feedback. Ethical review is underway to assess learnability, visual cueing, and embodiment risks.

Index terms: Mixed Reality, Sensory engagement, Adolescent health, Blue space, Wellbeing, Virtual nature, User Experience.

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1 INTRODUCTION

A European project called SOPHIE (Sea, Oceans, & Public Health in Europe) 2020 has identified a Strategic Research Agenda (SRA). This SRA will help inform Oceans and Human Health (OHH) research which investigates the complex links between human health and ocean health. The three main target areas are as follows: *Target 1. Sustainable Seafood and Healthy People*, *Target 2. Blue Spaces, Tourism and Well-being*, and *Target 3. Marine Biodiversity and Medicine*. The intended Target Area 2 envisions enhanced interactions with blue spaces to improve mental health and wellbeing of individuals and communities. As it is unrealistic to get everyone to the sea, SOPHIE has identified a gap in simulated blue environments, virtual reality or art-based experiences [3]. In 2018, the OECD (Organization for Economic Co-operation and Development) stated in a report that costs related to mental ill health were estimated to total EUR 600 billion across the European Union, while more than one in six people in European countries suffer from mental health problems [4]. This is a tremendous economic and public burden. Moreover, the World Health Organization (WHO) estimated that 77% of diseases in European states develop from non-communicable diseases (NCDs) which could be preventable [5]. Evidence suggests that activities in natural environments can improve mental wellbeing [6] while

virtual nature exposure may offer a promising alternative intervention [7].

2 RESEARCH GAP

Previous research in this area has predominantly focused on Virtual Reality (VR) experiences that mimic nature exposure through audio and visuals to relax the audience. Whereas this project highlights activities in a virtual blue space to boost the participants' wellbeing. The chosen target audience are adolescents and young adults, as this group is vulnerable to mental health disorders.

3 PROGRESS

The project aims to create an interactive blue space environment as an immersive experience. It exposes users to virtual nature, encourages exploration of the digital landscape, and facilitates learning a dance routine.

3.1 Location

This experience is set at Taputeranga Marine Reserve, a coastal site in south Wellington, Aotearoa New Zealand [8]. Selected for its rich biodiversity, changeable weather, and unique underwater landscape, the reserve features thriving kelp forests, diverse fish species, octopus, starfish, anemones, and sea sponges. Birdlife includes oystercatchers and albatrosses, alongside little blue penguins, and fur seals. Taputeranga Island, see Figure 1, stands out for its offshore position, untouched wildlife, and clear rock pools. Additional geological highlights appear at Ōwhiro Bay, Pariwhero, Red Rocks, and The Sirens Rocks, an intertidal zone rich with rock formations, tide pools, and abundant flora and fauna.

3.2 Representation

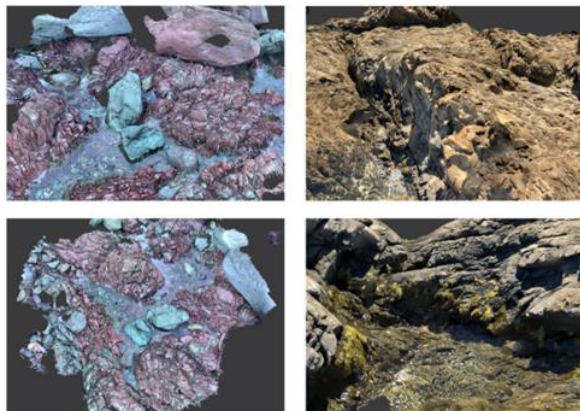


Figure 2: 3D scans of rock-formations and pools, left: Red Rocks, right: The Sirens Rocks.

The representation of users in this project is limited to hand representation and shadows of the body. The dancer that the audience meets is pictured through video footage to create a lifelike encounter. Communication between audience and performer is predominantly through music, sound cues, and eye contact or hand gestures.

To build the experience, this study utilizes a Meta Quest 3 headset set on passthrough mode with 6DoF (six degrees of freedom), headset tracking, and hand tracking. This allows the user to see the

virtual world overlaid the real world and interact with their surroundings without using controllers. To support this augmented content, a 270-degrees screen (CAVE) is employed to project a virtual backdrop which displays durational patterns such as daytime/night, and a variety of weather phenomena, see Figure 3.



Figure 3: Weather phenomena: cloudy sunset sky backdrop, real-world footage.

3.3 Interaction

The intention is to create positive ocean encounters that boost participants' wellbeing through three key elements:

1. Immersive Exploration: Participants trigger changes in the virtual water landscape using hand gestures, such as disturbing the water surface.
2. Dance Activity: Participants can observe a dancer demonstrating a routine, individually or in groups, and join when ready. This serves as both social encounter and physical exercise [9].
3. Blue Space Reconnection: The experience supports reconnection to real blue spaces, an underexplored area of virtual nature applications [10].

Located in gallery spaces, this immersive experience enables group viewing and engagement, supporting public health while highlighting human interdependence with nature. We also acknowledge opportunities to present this MR experience to landlocked audiences, those in healthcare settings, or individuals with limited mobility, for whom adjustments like isolating upper-body, head, or hand movements can be made.

3.4 Dance Exercise

The dance choreography was designed with a younger audience in mind. Simple, easy-to-follow movements, divided into hand, body, and leg segments, were developed in collaboration with a performance artist. Unlike exergames, this exercise follows a no-reward, non-gamified approach [11, 12]. In addition to supporting wellbeing through dance [13], the experience integrates breathing exercises across three phases:

1. Grounding Phase: Participants sit cross-legged, breathing deeply while synchronizing connected hand movements to anchor themselves spatially.
2. Upper Body Introduction: The performer slowly introduces dance with upper-body and arm motions, including hand clapping.

3. Full Elevation: The dancer performs elevated standing movements with stretches and coordinated hand clapping.

This structure engages participants in meaningful interaction with a water-based virtual environment that supports psychological wellbeing. We acknowledge that personal wellbeing varies by individual and depends on cultural context, prior experience [14], and existing place-based interactions or emotions [15]. Furthermore, this research critically examines how relationships with water are formed globally and how these influence blue space and wellbeing research [16, 17].



Figure 4: Dance choreography stages: a) cross-legged grounding with breathing and hand movements, b) upper-body introduction with hand clapping, c) wave-like arm movements, and d) elevated standing stretches with coordinated clapping.

3.5 Soundscape and Music

The soundscape comprises recorded natural sounds that create localized audio within the coastal 3D environment, enhancing spatial immersion at Taputeranga Marine Reserve. Complementing this, original music was developed in collaboration with a multidisciplinary sound artist, featuring wordless vocals alongside cues to guide the dancer's movements. These compositions draw inspiration from local bird calls, wind patterns, and wave rhythms, reinforcing the multi-sensory connection to ocean blue spaces while supporting the breathing and dance phases.

4 USER DESIGN STUDY

This research explores connected XR experiences designed to support young people's mental wellbeing through virtual blue space activities. Guiding research questions include:

1. How are relationships with ocean spaces formed in different world regions, and how can memories of these experiences be evoked in audiences?
2. What emotional responses emerge after exposure to virtual landscapes and dance exercises?
3. Do participants experience stress relief from exercise combined with virtual water exposure?

These questions will be addressed via participant workshops, with further investigation planned. A user design study, currently under ethical review, incorporates questionnaires on participants' emotional feedback, learnability [18], visual cueing [19], and embodiment risks such as cybersickness [20].

To accommodate diverse cultural backgrounds and upbringings, the study incorporates workshops in coastal locations, Wellington, Aotearoa New Zealand, and Busan, South Korea, in 2026. Participants may provide responses via writing, drawing, or other formats.

5 CONCLUSION

To summarize, the proposed experience engages young people in seaside movement activities to foster mental wellbeing through layered immersion, interaction, learning, and physical engagement. It progresses as follow, first, users enter a calming coastal soundscape and landscape. Second, they freely explore the environment. And third, they learn a simple dance routine guided by a performer with integrated breathing exercises. Success of this virtual water exposure will be evaluated through participatory workshops currently in development.

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FIGURE CREDITS

Figure 1: image credit: Nina Fern; Taputeranga Marine Reserve, Wellington, New Zealand, 2026.

Figure 2: image credit: Nina Fern; 3D scans of rock-formations and pools, left: Red Rocks, right: The Sirens Rocks, 2025.

Figure 3: image credit: Nina Fern; Weather phenomena: cloudy sunset sky backdrop, captured on location at Taputeranga Marine Reserve, 2025.

Figure 4: image credit: Nina Fern; Dance choreography stages: a) cross-legged grounding with breathing and hand movements, b) upper-body introduction with hand clapping, c) wave-like arm movements, and d) elevated standing stretches with coordinated clapping, 2026.

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